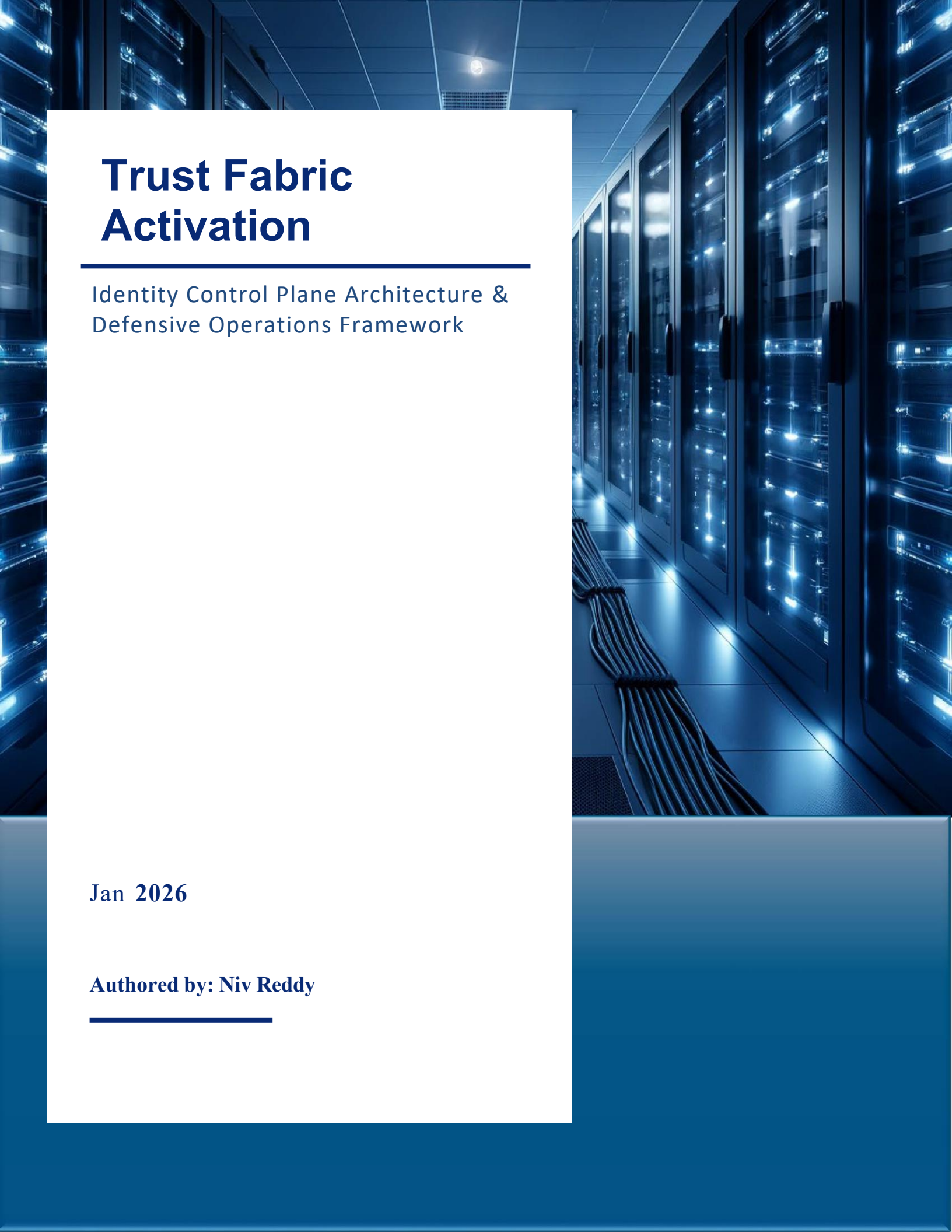




Trust Fabric Activation

Identity Control Plane Architecture &
Defensive Operations Framework

Jan 2026

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Introduction

Context

This stage activates routed trust relationships between previously sealed enterprise trust zones and onboards the first identity-governed workloads into the fabric.

This phase transitions the environment from:

- Physically isolated VLAN trust zones
- Into a routed, policy-enforced, identity-mediated enterprise security fabric

Segmentation is no longer only physical. It becomes policy-enforced, deterministic, and auditable.

All routing introduced in this phase is:

- Explicit
- Minimal
- Identity-centric
- Centrally enforced

No workload can communicate without passing through an identity-aware policy rail.

Trust Fabric Architecture

Trust Zone	VLAN	Role
Server Zone	10	Tier-0 and Tier-1 infrastructure
Workstation Zone	20	User endpoints
Security/Admin Zone	30	Security control plane
FW01 (pfSense)	Routed Broker	Inter-zone policy enforcement

Routed Trust Broker Deployment (FW01)

Purpose

Introduce a centralized policy-enforcing routing authority between enterprise trust zones

Implementation

A dedicated pfSense firewall was deployed as FW01

Interface	VLAN	Address
WAN	Home LAN	192.168.4.182/22
LAN	VLAN 10	10.0.10.1
OPT1	VLAN 20	10.0.20.1
OPT2	VLAN 30	10.0.30.1

vmbr1 now operates as a routed enterprise trunk

No trust zone can communicate without FW01 mediation.

Server Workload Attachment (FS01)

Attribute	Value
Hostname	FS01
VLAN	10
IP	10.0.10.20
Role	File Server

Identity Validation

FS01 was domain-joined and validated to:

- Authenticate using Kerberos
- Retrieve SYSVOL and Group Policy
- Resolve DNS exclusively through DC01
- Remain unreachable from VLAN 20 except through identity rails

FS01 becomes the first Tier-1 workload anchored to the identity fabric.

Workstation Workload Attachment (WS01)

Attribute	Value
Hostname	WIN10-01
VLAN	20
IP	10.0.20.10
Role	User Endpoint

Identity Validation

WS01 was domain-joined and validated to:

- Authenticate using Kerberos across routed trust rails
- Retrieve SYSVOL and Group Policy
- Resolve DNS only via DC01
- Remain fully isolated from Server Zone except identity ports
- Have zero lateral movement capability

This confirms that user endpoints are no longer flat-network clients. They are identity-gated workloads

Identity Rail Enforcement

FW01 enforces identity-only communication paths between VLAN 20 and VLAN 10 for now

Function	Ports
DNS	53 TCP/UDP
Kerberos	88 TCP/UDP
LDAP	389 TCP/UDP
SMB SYSVOL	445 TCP
RPC Mapper	135 TCP
Dynamic RPC	49152–65535 TCP
NTP	123 UDP
ICMP	Gateway only

Validation Results

Test	Result
Kerberos ticket issuance	Verified
SYSVOL and NETLOGON	Verified
DNS resolution	Verified
Lateral movement	Blocked
Cross-VLAN ping	Blocked
Identity-only access	Enforced